Meridian Health Group — Care Coordination Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Referral Service	Referral creation, urgency routing, admin assignment queue, FHIR ServiceRequest write-back	Lambda + PostgreSQL 15 + FHIR Adapter
Care Plan Service	Care plan CRUD, task assignment, review date alerts, FHIR CarePlan sync	Lambda + PostgreSQL 15 + FHIR Adapter
Check-In Service	Walk-in queue management, arrival status, no-show prompting, FHIR Appointment write-back	Lambda + DynamoDB (real-time queue state)
Waitlist Service	Waitlist registration, slot-offer workflow, 30-minute response window	Lambda + DynamoDB + Notification Service
Admin Service	Proxy access approval workflow, Cognito grant management, proxy revocation	Lambda + DynamoDB + Cognito

Frontends Owned by This Team

Client	Technology	Notes
Clinician Portal	React 18 SPA (TypeScript) + GraphQL subscriptions	Role-gated by Okta/Cognito claims (GP / Specialist / Nurse / Admin); real-time inbox badge via GraphQL subscription
Front Desk Display	React 18 SPA (kiosk mode, full-screen)	Walk-in queue and appointment schedule; auto-refreshes every 30 seconds; displayed on reception screen
Waiting Room Display	React 18 SPA (read-only, public)	Queue position and estimated wait time; no PHI displayed (only queue number and estimated wait)

Key Technical Constraints

Real-Time Inbox & GraphQL Subscriptions

- The clinician inbox uses **GraphQL subscriptions** (AWS AppSync WebSocket) for real-time badge updates
- When a new message, prescription request, or referral arrives, an EventBridge event triggers an AppSync mutation, which pushes an update to all subscribed clinicians
- Urgency escalation (unread urgent message after 30 minutes) is a scheduled Lambda that checks DynamoDB and triggers a Twilio SMS to the specialist's supervisor

FHIR Adapter — Care Coordination Context

- All FHIR reads from the Clinician Portal use **SMART on FHIR context scoping** — the FHIR Adapter enforces that the requesting clinician has a valid care relationship with the patient before returning PHI
- Care relationship validation: the clinician's `practitionerId` (from JWT claim) is checked against the `Appointment.participant` or `CarePlan.author` list for the patient in Epic
- Write-back latency target: FHIR writes (referral creation, appointment status, care plan update) must confirm within **3 seconds** of clinician action

Walk-In Queue (Real-Time State)

- Walk-in queue state stored in DynamoDB (not PostgreSQL) for low-latency real-time reads
- Front Desk Display refreshes every 30 seconds (polling); Walk-in patients added to the queue immediately on check-in
- Queue position estimation: average consultation time (rolling 60-minute average per clinician) used to calculate estimated wait; displayed on the Waiting Room screen
- No PHI on the Waiting Room Display — only queue number and estimated wait time (patient given their queue number on arrival)

Proxy Access Approval

- Proxy request documents (scanned consent forms) uploaded to S3 via pre-signed URL; ClamAV virus-scanned before display to admin
- Admin approval action: calls the Admin Service → Auth Service → Cognito `adminUpdateUserAttributes` to set `custom:proxyFor` attribute
- Approval is synchronous from admin's perspective; Cognito update confirmed before success response returned

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External Integrations (this team)

Integration	Protocol	Direction	Notes
FHIR Adapter → Epic	FHIR R4 REST (SMART on FHIR)	Referral/CarePlan/CheckIn Service → FHIR Adapter → Epic	Referral ServiceRequest, CarePlan, Appointment status write-backs; patient record reads
AWS AppSync	WebSocket (GraphQL subscription)	Clinician Portal ← AppSync	Real-time inbox badge; new message/referral/prescription notifications
AWS Cognito	SDK	Admin Service ↔ Cognito	Proxy access grant management; role claim updates
Twilio	REST API	Notification Service → Twilio	Urgent referral escalation SMS; waitlist slot offer SMS
SendGrid	REST API	Notification Service → SendGrid	Referral confirmation email to patient; appointment confirmed email
ClamAV Lambda	S3 event trigger	S3 → ClamAV Lambda	Virus scan proxy consent form before admin review
EventBridge	Event bus	Care Coordination Services → EventBridge	`referral.created`, `appointment.arrived`,

			`careplan.task.completed` events

Non-Functional Requirements (this team)

Category	Requirement
FHIR write-back	Referral, care plan, and appointment status writes confirm within 3 seconds
Inbox real-time update	New inbox item pushed to clinician within 5 seconds of EventBridge event via AppSync subscription
Urgent escalation	Unread urgent message triggers supervisor SMS within 30 minutes of delivery
Walk-in queue refresh	Front Desk Display refreshes every 30 seconds; queue state accurate to within 30 seconds
SMART on FHIR enforcement	FHIR Adapter returns 403 for any request where the clinician does not have a verified care relationship with the patient
Proxy approval latency	Cognito attribute update confirmed synchronously; admin sees "approved" confirmation within 3 seconds
HIPAA audit	All PHI access events in the clinician portal emitted to Audit Service within 1 second (EventBridge async)